



UNIVERSITAS GADJAH MADA

Introduction to Image Processing



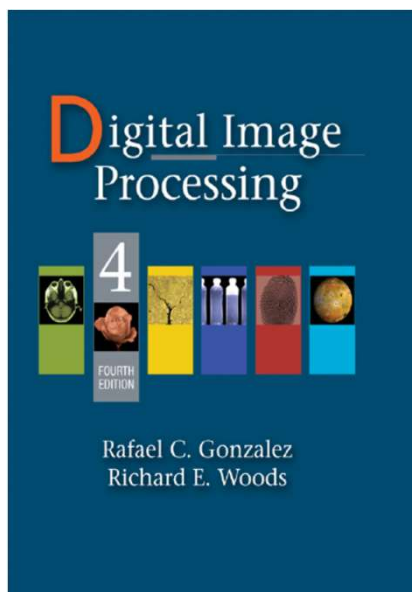
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Text book and Evaluation Components

- Text book
 - Digital Image Processing, R.C. Gonzalez & R.E. Woods, Pearson [2018]
- Evaluation Components
 - Assignments
 - Quizes
 - Midterm Exam
 - Final Exam

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Objectives

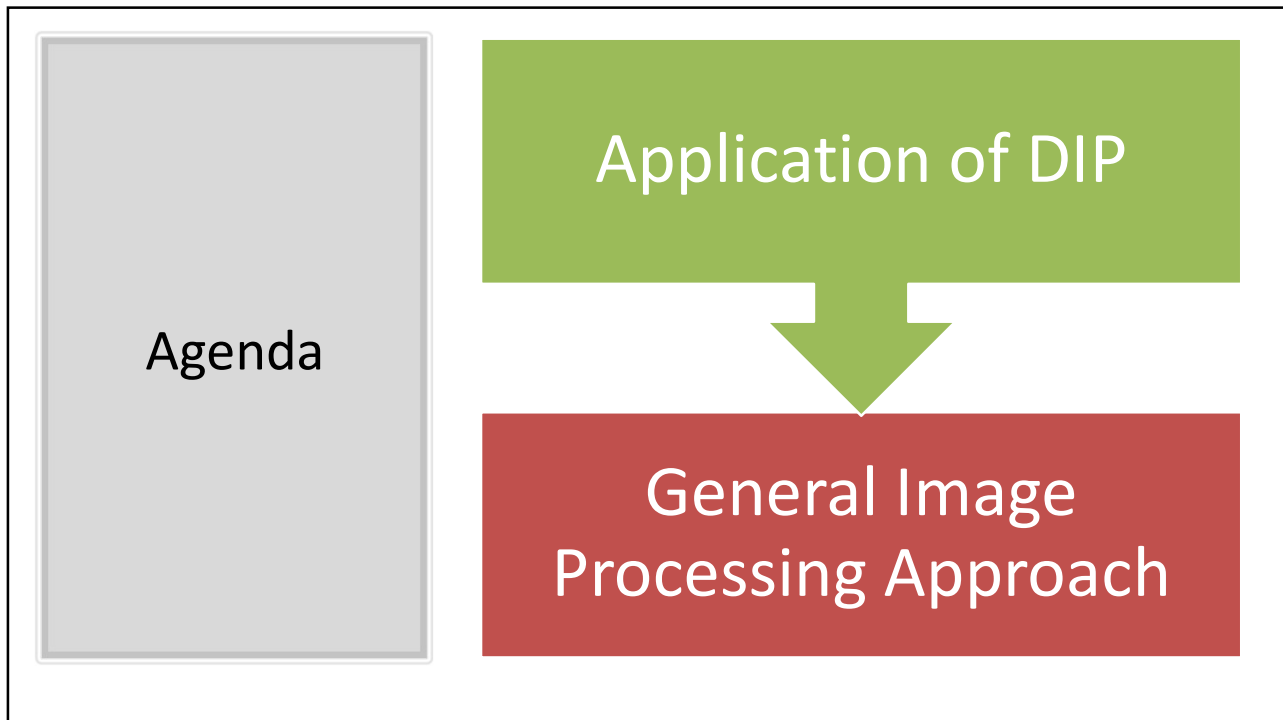


- Understand the concept of a digital image.
- Have a broad overview of the historical underpinnings of the field of digital image processing.
- Understand the definition and scope of digital image processing.
- Know the fundamentals of the electromagnetic spectrum and its relationship to image generation.
- Be aware of the different fields in which digital image processing methods are applied.
- Be familiar with the basic processes involved in image processing.
- Be familiar with the components that make up a general-purpose digital image processing system.
- Be familiar with the scope of the literature where image processing work is reported.

Topics



- Basic concepts of
 - Image enhancement
 - Image restoration
 - Image analysis
- Mathematical tools
 - Image pixel value transformation (e.g. histogram processing)
 - Image filtering (linear, nonlinear)
 - Image transformation (DFT, DWT)
 - Image pixel relations and manipulation
 - Morphological processing
 - Statistical inference: classification, clustering
 - Etc.

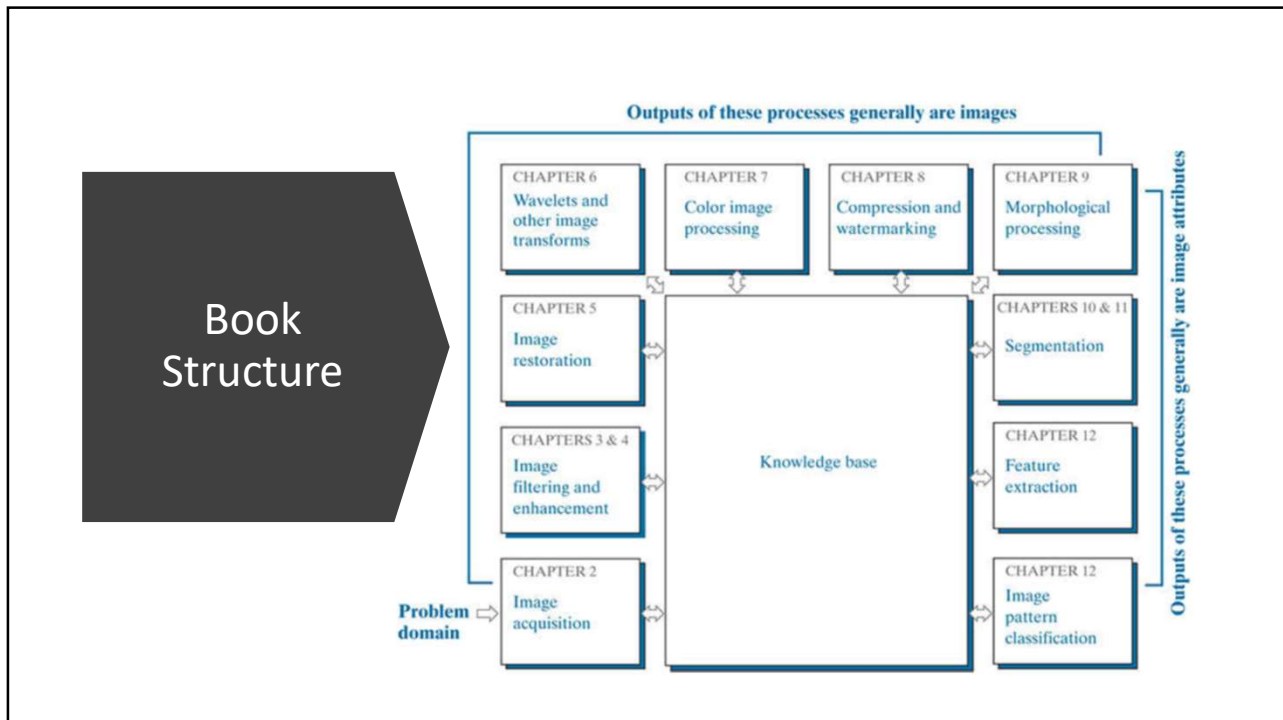


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Application Examples

- Lab. Elins channel
<https://www.youtube.com/channel/UC11p6PdFol4VefKH6JFDXuA>
- Semilin: Menuju Indonesia Merdeka Teknologi (menit ke 1.05.40 – 2.14.00)
<https://www.youtube.com/watch?v=7Uwgr9Miy1w>

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Text Book Coverage



- Image acquisition
 - Obtaining a digital image
 - May include preprocessing, such as scaling
- Image Filtering and Enhancement
 - manipulating an image to be more suitable for a specific application.
 - problem oriented, subjective
- Image restoration
 - improving the appearance of an image
 - image restoration is objective, tend to be based on mathematical or probabilistic models of image degradation

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Text Book Coverage



- Wavelets and other image transforms
 - *Wavelets* are foundation for representing images in various degrees of resolution, used for data compression, and for pyramidal representation
 - Other transforms, include Fourier transform
- Color Image Processing
 - color models and basic color processing
 - basis for extracting features of interest in an image.
- Compression and Water marking
 - techniques for reducing the storage required to save an image, or the bandwidth required to transmit it
 - Watermarking: embedding information on the image

Text Book Coverage



- *Morphological processing*
 - tools for extracting image components useful in the representation and description of shape
 - begins a transition from processes that output images to processes that output image attributes
- *Segmentation*
 - partitions an image into its constituent parts or objects
 - autonomous segmentation is difficult tasks in image processing
 - Leads to solution that require objects to be identified individually
 - accurate segmentation leads to successful automated object classification

Text Book Coverage



- *Feature extraction*
 - almost always follows the output of a segmentation stage
 - consists of feature detection and feature description
 - *Feature detection* refers to finding the features in an image, region, or boundary
 - *Feature description* assigns quantitative attributes to the detected features
 - Example: detect corners in a region, and describe those corners by their orientation and location
 - subdivided into three, whether they are applicable to boundaries, regions, or whole images
 - Feature descriptors should be insensitive to variations in parameters (scale, translation, rotation, illumination, and viewpoint)

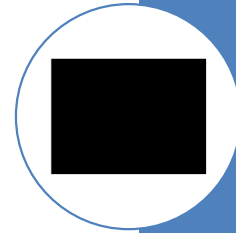
Text Book Coverage



- *Image pattern classification*
 - labels an object based on its feature descriptors (e.g., “vehicle”)
 - “classical” approaches:
 - *Minimum distance,*
 - *correlation,*
 - *Bayes classifiers, etc*
 - modern approaches:
 - *deep neural networks,*
 - *deep convolutional neural networks*

This course

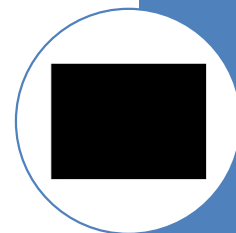
- Pre-processing
- Segmentation
- Feature Extraction
- Feature Selection
- Feature Matching



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Pre-processing

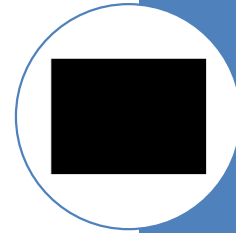
- Enhancement
 - Noise reduction
 - Space transformation
- Normalization
 - Size normalization
 - Intensity normalization



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Segmentation

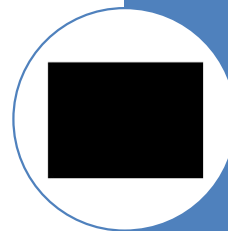
- Splits data into segments
- RoI (contains single object)



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Feature extraction

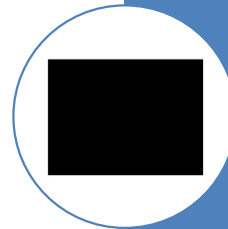
- Features depend on data
- Features
 - Distinguishing
 - Robust
 - Small in numbers
 - Non-correlated
- Examples
 - Term, term frequency, term weight
 - Spectrum, energy distribution
 - Color, shape, topological, texture features



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Feature selection

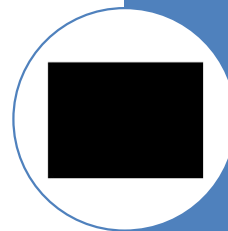
- Not too few, but not too much
- Some features are more important than others
- Needs to select significant features
 - PCA
 - Decision tree
 - etc



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Feature matching

- Similarity measures
 - Cosine coefficient
- Distance measures
 - Euclidean distance
 - Mahalanobis distance
- Structural properties
- Correlation
- String matching



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**Terimakasih
Maturnuwun**

THANK YOU



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